



# Surds and Indices

## CONCEPT

$$(1024)^{1/10} = \left(\frac{1}{1024}\right)^{1/10} = \left(\frac{1}{2^{10}}\right)^{1/10} = \frac{1}{2^{10 \times \frac{1}{10}}} = \frac{1}{2} = \underline{\underline{\frac{1}{2}}}$$

### 1. Laws of Indices:

i.  $a^m \times a^n = a^{m+n}$

ii.  $\frac{a^m}{a^n} = a^{m-n}$

iii.  $(a^m)^n = a^{mn}$

iv.  $(ab)^n = a^n b^n$

v.  $\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$

vi.  $a^0 = 1$   $a^{-1} = \frac{1}{a}$

### 2. Surds:

Let  $a$  be a rational number and  $n$  be a positive integer such that  $a^{(1/n)} = \sqrt[n]{a}$

Then,  $\sqrt[n]{a}$  is called a surd of order  $n$ .

### 3. Laws of Surds:

i.  $\sqrt[n]{a} = a^{(1/n)}$

ii.  $\sqrt[n]{ab} = \sqrt[n]{a} \times \sqrt[n]{b}$

iii.  $\sqrt[n]{\frac{a}{b}} = \frac{\sqrt[n]{a}}{\sqrt[n]{b}}$

iv.  $(\sqrt[n]{a})^n = a$

v.  $\sqrt[m]{\sqrt[n]{a}} = \sqrt[mn]{a}$

vi.  $(\sqrt[n]{a})^m = \sqrt[n]{a^m}$

3. Given that  $10^{0.48} = x$ ,  $10^{0.70} = y$  and  $x^z = y^2$ , then the value of  $z$  is close to:

A. 1.45

B. 1.88

C. 2.9

D. 3.7

$$10^{0.48z} = 10^{0.7 \times 2}$$

$$0.48z = 1.4$$

$$z = \frac{1.4}{0.48} = \frac{140}{48} \approx \underline{\underline{2.9}}$$

9.  $(25)^{7.5} \times (5)^{2.5} \div (125)^{1.5} = 5^?$

A. 8.5

☒ B. 13

C. 16

D. 17.5

E. None of these

$$5^{2 \times 7.5} \times 5^{2.5} \div 5^{3 \times 1.5} = 5^x$$

$$15 + 2.5 - 4.5 = x$$

$$x = \underline{\underline{13}}$$

10.  $(0.04)^{-1.5} = ?$

A. 25

☒ B. 125

C. 250

D. 625

$$\left(\frac{4}{100}\right)^{-3/2} = \left(\frac{100}{4}\right)^{3/2} = 25^{3/2}$$

$$= 5^{2 \times 3/2}$$

$$= 5^3 = \underline{\underline{125}}$$

8.  $\frac{1}{1+x(b-a)+x(c-a)} + \frac{1}{1+x(a-b)+x(c-b)} + \frac{1}{1+x(b-c)+x(a-c)} = ?$

A. 0

☒ B. 1

C.  $x^{a-b-c}$

D. None of these

$$1 + \frac{x^b}{x^a} + \frac{x^c}{x^a}$$

$$= \frac{x^a + x^b + x^c}{x^a + x^b + x^c} = \underline{\underline{1}}$$

